

CS 171 / EE 142: Intro to ML and DM

Christian Shelton

UC Riverside

Slide Set 2: Notation and Bayes Optimality



- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
- DO NOT REDISTRIBUTE
 - ▶ For use only by enrolled students in the course

Supervised Learning, Example

The Data (training set):

$$\mathbf{X} = \begin{array}{c} \text{sepal length (cm)} \quad \text{sepal width (cm)} \quad \text{petal length (cm)} \quad \text{petal width (cm)} \\ \left[\begin{array}{cccc} 6.3 & 2.3 & 4.4 & 1.3 \\ 6.4 & 2.7 & 5.3 & 1.9 \\ 5.4 & 3.7 & 1.5 & 0.2 \\ 6.1 & 3.0 & 4.6 & 1.4 \\ 5.0 & 3.3 & 1.4 & 0.2 \\ \vdots & \vdots & \vdots & \vdots \\ 4.9 & 3.1 & 1.5 & 0.1 \end{array} \right] \end{array} \quad \mathbf{Y} = \begin{array}{c} \text{class} \\ \left[\begin{array}{c} \text{versicolor} \\ \text{virginica} \\ \text{setosa} \\ \text{versicolor} \\ \text{setosa} \\ \vdots \\ \text{setosa} \end{array} \right] \end{array}$$

Supervised Learning, Example

The Data (training set):

$$\mathbf{X} = \begin{array}{ccccc} & \text{sepal length (cm)} & \text{sepal width (cm)} & \text{petal length (cm)} & \text{petal width (cm)} \\ \left[\begin{array}{c} 6.3 \\ 6.4 \\ 5.4 \\ 6.1 \\ 5.0 \\ \vdots \\ 4.9 \end{array} \right. & & & & \end{array} \quad \mathbf{Y} = \begin{array}{c} \text{class} \\ \left[\begin{array}{c} 1 \\ 2 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{array} \right] \end{array}$$

Supervised Learning, Example

The Data (training set):

$$\mathbf{X} = \begin{bmatrix} 6.3 & 2.3 & 4.4 & 1.3 \\ 6.4 & 2.7 & 5.3 & 1.9 \\ 5.4 & 3.7 & 1.5 & 0.2 \\ 6.1 & 3.0 & 4.6 & 1.4 \\ 5.0 & 3.3 & 1.4 & 0.2 \\ \vdots & \vdots & \vdots & \vdots \\ 4.9 & 3.1 & 1.5 & 0.1 \end{bmatrix}$$

$$\mathbf{Y} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Supervised Learning, Example

The Data (training set):

$$\mathbf{X} = \left[\begin{array}{cccc} 6.3 & 2.3 & 4.4 & 1.3 \\ 6.4 & 2.7 & 5.3 & 1.9 \\ 5.4 & 3.7 & 1.5 & 0.2 \\ 6.1 & 3.0 & 4.6 & 1.4 \\ 5.0 & 3.3 & 1.4 & 0.2 \\ \vdots & \vdots & \vdots & \vdots \\ 4.9 & 3.1 & 1.5 & 0.1 \end{array} \right] \left. \vphantom{\begin{array}{c} \\ \\ \\ \\ \\ \\ \end{array}} \right\} m$$

$\underbrace{\hspace{10em}}_n$

$$\mathbf{Y} = \left[\begin{array}{c} 1 \\ 2 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{array} \right]$$

Supervised Learning, Example

The Data (training set):

$$\mathbf{X} = \begin{bmatrix} 6.3 & 2.3 & 4.4 & 1.3 \\ 6.4 & 2.7 & 5.3 & 1.9 \\ 5.4 & 3.7 & 1.5 & 0.2 \\ 6.1 & 3.0 & 4.6 & 1.4 \\ 5.0 & 3.3 & 1.4 & 0.2 \\ \vdots & \vdots & \vdots & \vdots \\ 4.9 & 3.1 & 1.5 & 0.1 \end{bmatrix} \quad \vec{x}_2^T$$

$$\mathbf{Y} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad y_2$$

Supervised Learning, Example

The Data (training set):

$$\mathbf{X} = \begin{bmatrix} 6.3 & 2.3 & 4.4 & 1.3 \\ 6.4 & 2.7 & 5.3 & 1.9 \\ 5.4 & 3.7 & 1.5 & 0.2 \\ 6.1 & 3.0 & 4.6 & 1.4 \\ 5.0 & 3.3 & 1.4 & 0.2 \\ \vdots & \vdots & \vdots & \vdots \\ 4.9 & 3.1 & 1.5 & 0.1 \end{bmatrix}$$

Annotations:

- A blue box highlights the first row of $\mathbf{X}$ (excluding the first column), labeled $\vec{x}_2^T$.
- A blue box highlights the element 3.3 in the fifth row and second column, labeled $x_{5,2}$.

$$\mathbf{Y} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Annotation:

- A blue box highlights the element 2 in the second row, labeled y_2 .

Supervised Learning, Example

The Data (training set):

$$\mathbf{X} = \begin{bmatrix} 6.3 & 2.3 & 4.4 & 1.3 \\ 6.4 & 2.7 & 5.3 & 1.9 \\ 5.4 & 3.7 & 1.5 & 0.2 \\ 6.1 & 3.0 & 4.6 & 1.4 \\ 5.0 & 3.3 & 1.4 & 0.2 \\ \vdots & \vdots & \vdots & \vdots \\ 4.9 & 3.1 & 1.5 & 0.1 \end{bmatrix}$$

Annotations: A blue box highlights the second row of $\mathbf{X}$, labeled $\vec{x}_2^T$. A blue box highlights the value 3.3 in the second column of the fifth row, with a line pointing to it labeled $x_{5,2}$.

$$\mathbf{Y} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Annotation: A blue box highlights the value 2 in the second row of $\mathbf{Y}$, labeled y_2 .

A general example:

$$\vec{x} = \begin{bmatrix} 5.8 \\ 2.9 \\ 4.7 \\ 1.5 \end{bmatrix}$$

Annotation: A blue box highlights the value 2.9 in the second row of $\vec{x}$, labeled x_2 .

$$y = 1$$

Probabilistic Assumptions

In machine learning we (often) assume

- The data are drawn at random

Probabilistic Assumptions

In machine learning we (often) assume

- The data are drawn at random
- The training data and the testing data come from the same distribution

Probabilistic Assumptions

In machine learning we (often) assume

- The data are drawn at random
- The training data and the testing data come from the same distribution
- Each element of data ($\vec{x}, y$ pair) is drawn independently from the same distribution (known as iid)

Probabilistic Assumptions

In machine learning we (often) assume

- The data are drawn at random
- The training data and the testing data come from the same distribution
- Each element of data ($\vec{x}, y$ pair) is drawn independently from the same distribution (known as iid)
- This distribution, $p(\vec{x}, y)$, is unknown

Probabilistic Assumptions

$$\mathbf{X} = \begin{bmatrix} 6.3 & 2.3 & 4.4 & 1.3 \\ 6.4 & 2.7 & 5.3 & 1.9 \\ 5.4 & 3.7 & 1.5 & 0.2 \\ 6.1 & 3.0 & 4.6 & 1.4 \\ 5.0 & 3.3 & 1.4 & 0.2 \\ \vdots & \vdots & \vdots & \vdots \\ 4.9 & 3.1 & 1.5 & 0.1 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\left(\begin{bmatrix} 6.3 \\ 2.3 \\ 4.4 \\ 1.3 \end{bmatrix}, 1 \right) \sim p(\vec{x}, y)$$

Probabilistic Assumptions

$$\mathbf{X} = \begin{bmatrix} 6.3 & 2.3 & 4.4 & 1.3 \\ 6.4 & 2.7 & 5.3 & 1.9 \\ 5.4 & 3.7 & 1.5 & 0.2 \\ 6.1 & 3.0 & 4.6 & 1.4 \\ 5.0 & 3.3 & 1.4 & 0.2 \\ \vdots & \vdots & \vdots & \vdots \\ 4.9 & 3.1 & 1.5 & 0.1 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\left(\begin{bmatrix} 6.4 \\ 2.7 \\ 1.5 \\ 1.9 \end{bmatrix}, 2 \right) \sim p(\vec{x}, y)$$

Probabilistic Assumptions

$$\mathbf{X} = \begin{bmatrix} 6.3 & 2.3 & 4.4 & 1.3 \\ 6.4 & 2.7 & 5.3 & 1.9 \\ 5.4 & 3.7 & 1.5 & 0.2 \\ 6.1 & 3.0 & 4.6 & 1.4 \\ 5.0 & 3.3 & 1.4 & 0.2 \\ \vdots & \vdots & \vdots & \vdots \\ 4.9 & 3.1 & 1.5 & 0.1 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\left(\begin{bmatrix} 5.4 \\ 3.7 \\ 1.5 \\ 0.2 \end{bmatrix}, 0 \right) \sim p(\vec{x}, y)$$

Why a Distribution

$$[6.4 \quad 3.2 \quad 4.5 \quad 1.5]^T, 1 \sim p(\vec{x}, y)$$

$$p(\vec{x} = [6.4 \quad 3.2 \quad 4.5 \quad 1.5]^T, y = 1)$$

can be viewed as

Why a Distribution

$$[6.4 \ 3.2 \ 4.5 \ 1.5]^T, 1 \sim p(\vec{x}, y)$$

$$p(\vec{x} = [6.4 \ 3.2 \ 4.5 \ 1.5]^T, y = 1)$$

can be viewed as

$$= p(\vec{x} = [6.4 \ 3.2 \ 4.5 \ 1.5]^T) p(y = 1 \mid \vec{x} = [6.4 \ 3.2 \ 4.5 \ 1.5]^T)$$

(chance we find a flower of this size times the chance a flower of this size is of type Versicolor)

Why a Distribution

$$[6.4 \ 3.2 \ 4.5 \ 1.5]^T, 1 \sim p(\vec{x}, y)$$

$$p(\vec{x} = [6.4 \ 3.2 \ 4.5 \ 1.5]^T, y = 1)$$

can be viewed as

$$= p(\vec{x} = [6.4 \ 3.2 \ 4.5 \ 1.5]^T) p(y = 1 \mid \vec{x} = [6.4 \ 3.2 \ 4.5 \ 1.5]^T)$$

(chance we find a flower of this size times the chance a flower of this size is of type Versicolor)

or

$$= p(y = 1) p(\vec{x} = [6.4 \ 3.2 \ 4.5 \ 1.5]^T \mid y = 1)$$

(chance we find a flower of type Versicolor times the chance a Versicolor has this size)

Why a Distribution

$$[6.4 \ 3.2 \ 4.5 \ 1.5]^T, 1 \sim p(\vec{x}, y)$$

$$p(\vec{x} = [6.4 \ 3.2 \ 4.5 \ 1.5]^T, y = 1)$$

can be viewed as

$$= p(\vec{x} = [6.4 \ 3.2 \ 4.5 \ 1.5]^T) p(y = 1 \mid \vec{x} = [6.4 \ 3.2 \ 4.5 \ 1.5]^T)$$

(chance we find a flower of this size times the chance a flower of this size is of type Versicolor)

or

$$= p(y = 1) p(\vec{x} = [6.4 \ 3.2 \ 4.5 \ 1.5]^T \mid y = 1)$$

(chance we find a flower of type Versicolor times the chance a Versicolor has this size)

Which one depends on how we view the “generation process,” but it doesn't matter.
Both demonstrate that there is a probabilistic relationship between x and y .

Testing Distribution

Performance of the resulting rule on $p(\vec{x}, y)$ is what we really care about.
Test set is just a proxy of the true distribution $p(\vec{x}, y)$.

Bayes-Optimal Classification

What if $p(\vec{x}, y)$ were known
and y were categorical?

Bayes-Optimal Classification

What if $p(\vec{x}, y)$ were known
and y were categorical?

What would be the best rule? (to minimize the chance of misclassification)

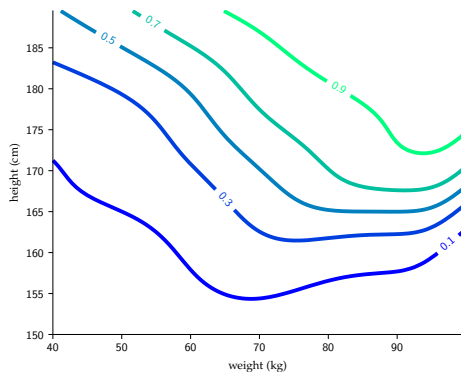
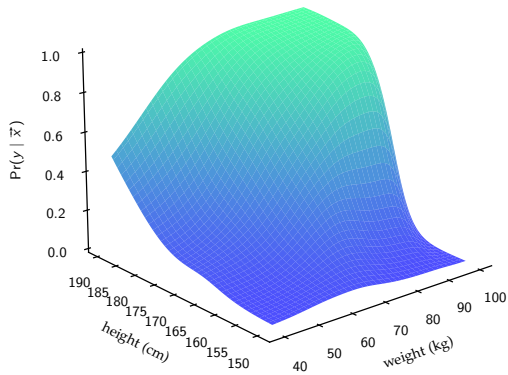
Bayes-Optimal Classification

What if $p(\vec{x}, y)$ were known
and y were categorical?

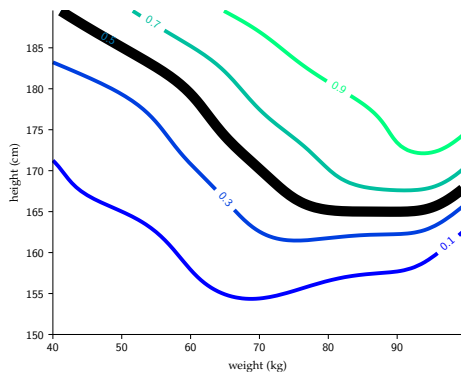
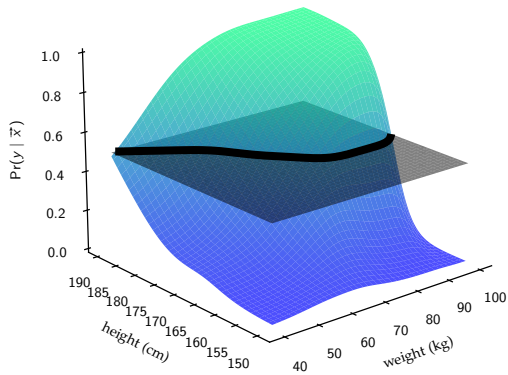
What would be the best rule? (to minimize the chance of misclassification)

$$f(\vec{x}) = \arg \max_c p(y = c \mid \vec{x})$$

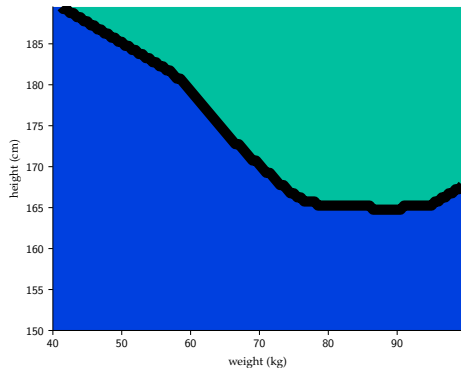
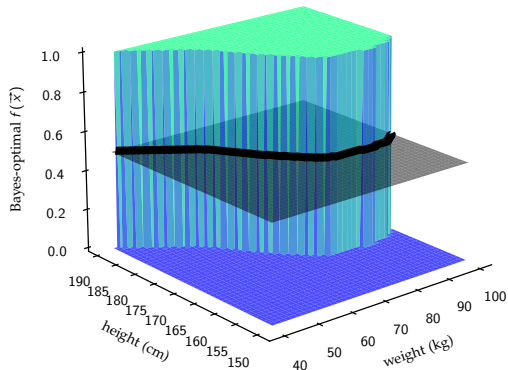
Two-class Bayes-Optimal Classification



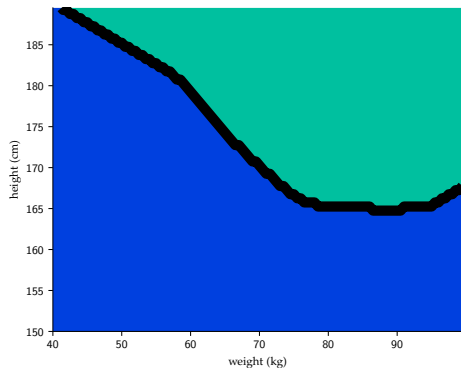
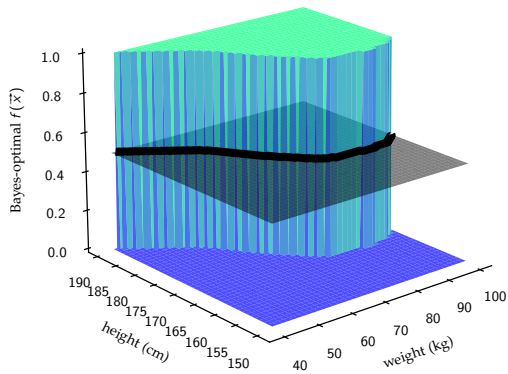
Two-class Bayes-Optimal Classification



Two-class Bayes-Optimal Classification



Two-class Bayes-Optimal Classification



Of course, $p(\vec{x}, y)$ or $p(y | \vec{x})$ are not known.
This (the Bayes-optimal classifier) is an ideal, but not generally achievable.